National Institute of Technology Warangal

CS26001: Advanced Algorithms - Course Project

Prediction of Imagined Words Using Brain EEG Signals

Primary Paper Rank-GA Baseline and Multi-Scale Riemannian Consensus (MDRC-AGA)

Multi-Scale 13-Band + Speech-10 Manifold with Dual-Protocol Leakage Audit

and Offline Oscilloscope Inspection

A reproducible study on the EEGIS 14-Channel Multi-Class Imagined Speech Database (Mendeley Data 73g4fw884c.1)

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Abstract

Imagined speech electroencephalography (EEG) offers a non-invasive communication pathway for individuals with severe motor impairments such as locked-in syndrome, brainstem stroke, and ALS. Across 4,044 Mendeley EEGIS trials (337 continuous 8s blocks, 128 Hz, 14 channels, 48-sample/37.5% chunk overlap, unrecoverable participant IDs), naive chunk splits inflate kNN accuracy via within-block overlap. Excluding all 9 test_signals/ blocks (108 chunks) yields N_ref=3,936 (328 blocks). Under leakage-free Block-Disjoint Nested 5-CV, all methods sit near 9-class chance (11.11%): Stage 1 Paper Rank-GA+kNN scores 10.01% (Yes/No 45.11%), True Tangent LogReg 10.72%, EEGNet Fold0 11.24%, and PCA+Calibrated RBF-SVM 11.41% (ECE=0.022).

Under the exploratory Overlapping-Chunk 5-CV protocol matching Lara-Arellano et al. (MDPI Eng, 2025; 96.0% published Yes/No), Stage 1 Paper 6B Rank-GA+kNN reaches 90.02% 9-class and 96.11% Yes/No. Stage 2 MDRC-AGA (1,695D three-scale Log-Euclidean SPD manifold: Mendeley 6B 630D + Fine 7B 735D + Speech-10 330D; MultiScaleBlockAGA; PCA-LDA + Platt-Ridge Stacker & [3:2:1:1] Consensus) reaches 91.84% (0.9180 F1, Brier 0.156, ECE 0.037) on Chunk 5-CV (95.67% Yes/No) and 9.68% Stacker (ECE=0.100) / 10.14% Vote on Block-Disjoint 5-CV. Sibling-excluded offline test_signals/ verify honest 0/9 generalization at 31.8 ms/epoch CPU compute.

Keywords: brain-computer interface (BCI); imagined speech; electroencephalography (EEG); multi-scale Riemannian covariance; adaptive genetic algorithm; manifold consensus; EEGIS dataset.

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1 Introduction

Brain-computer interfaces (BCIs) explore non-invasive communication pathways for individuals with severe motor impairments such as ALS or brainstem stroke whose motor tracts are disrupted while cognition remains intact. Unlike gaze-dependent P300 or SSVEP paradigms, imagined speech analyzes scalp EEG correlates of internal phonological planning across fronto-temporal speech-related channels and sensorimotor leads without audible sound or physical movement.

Building on Lara-Arellano et al. (Eng MDPI 2025; 96.0% binary Yes/No under exploratory chunk splits), we implement Stage 1 (Paper Rank-GA + kNN over 6-Band SPD covariance) and Stage 2 MDRC-AGA with equal-emphasis Dual-Protocol auditing across Leakage-Safe Block-Disjoint 5-CV (N_ref=3,936) and Exploratory Chunk 5-CV:

- Contribution 1: Multi-Scale 13-Band + Speech-10 SPD Manifold (1,695 Features): Fuses Mendeley 6-Band SPD (630D), Fine 7-Band sub-rhythms (735D, 4-45 Hz), and Speech-10 scalp montage (330D).
- Contribution 2: Fold-Safe Tangent Fit, MultiScaleBlockAGA & PCA-LDA/Ridge Stacker: Precomputes trial SPDs, fits C_ref only per train fold, evolves w ([0.85,0.66,1.49] blk; [1.30,0.98,0.72] chk).
- Contribution 3: Dual-Protocol Evaluation with Calibration (Brier, ECE, 95% CI): Chunk 5-CV: Stage 1 90.02% -> Stage 2 91.84% Vote (ECE 0.037) | Block-Disjoint: ~10.14%-11.41% (chance 11.11%).
- Contribution 4: Forensic Audit, Leakage Remediation & Offline Compute Bench (31.8 ms): Discloses 48-sample overlap, excludes all 108 test_signals/ sibling chunks (0/9 honest), and adds 8 unit tests.

2 Primary Research Paper & Related Work

2.1 Primary Reference Paper: Method, GA Design & Published Accuracy

Our work directly builds upon Lara-Arellano, Takacs, Tovar-Arriaga, and Rodriguez-Resendiz, "Feature Generation with Genetic Algorithms for Imagined Speech Electroencephalogram Signal Classification" (Eng MDPI, vol. 6, no. 4, art. 75, Apr. 2025; DOI: 10.3390/eng6040075). Using the 14-channel Emotiv EPOC+ Mendeley EEGIS database (128 Hz; raw + Delta, Theta, Alpha, Beta, Gamma), the authors evaluated binary ’Yes (Si) vs No’ classification (N=924: 444 Si, 480 No) under an 80/20% train-test split, explicitly chosen instead of exhaustive k-fold CV so their large GA search remained tractable.

Methodology Used: Each EEG epoch yielded 214 statistical and spectral features across the 6 bands, encoded alongside 14 channel-activation bits into a 228-bit binary chromosome (P=100 individuals, G=100 generations). Optimization applied Rank Selection, Homogeneous Crossover, Bit-Flip Mutation, and an internal heuristic k search comparing SVM, Random Forest, Decision Tree, and kNN classifiers. Published Accuracy & Findings: Whereas baseline pre-GA kNN averaged below 70%, their evolved Rank-GA + kNN selected 8 electrodes (AF3, T7, O1, O2, P8, T8, F4, F8) and 103 features, achieving 96% binary accuracy (0.97 Recall, 0.94 F1-score). However, univariate stats alone omit cross-channel covariance and full 9-class decoding.

Table 1: Methodological profile and published accuracy of the primary reference paper (Lara-Arellano et al., 2025).

Primary Paper Attribute (Eng 2025)	Published Specification / Reported Result
Reference & Task Scope	Lara-Arellano et al. (DOI: 10.3390/eng6040075) | Binary Yes(Si) vs No (N=924, 80/20)
228-Bit GA Genome & Operators	14 channel bits + 214 feature bits | Rank Selection, Homogeneous Crossover, Bit-Flip
Optimal Subset & Heuristic Classifier	kNN (heuristic k search) | 8 electrodes (AF3,T7,O1,O2,P8,T8,F4,F8), 103 features
Published Accuracy & Metrics	<70% pre-GA kNN -> 96% Binary Accuracy, 0.97 Recall, 0.94 F1 (binary only)

2.2 Comparative Related Literature

CSP, CNN & Riemannian BCIs: Under strict Block-Disjoint Nested 5-CV, FBCSP+SVM (Blankertz et al., 2011; 9.96%), scalar 6-band stats (10.34% SVM / 10.16% kNN7), MDM (8.74%), and 40-epoch EEGNetAttention (Lawhern et al., 2018; 11.24% Fold 0) all remain near 9-class chance (11.11%). Under Exploratory Chunk 5-CV matching Lara-Arellano et al., SPD tangent geometry enables Stage 1 (90.02% / 96.11% Yes/No) and Stage 2 Vote (91.84% / 95.67% Yes/No).

3 Methodology

3.1 Overview

Figure 1 contrasts Stage 1 (Paper 6B Rank-GA + kNN: 10.01% Blk-Disj / 90.02% Chunk) against Stage 2 MDRC-AGA (1,695D Three-Scale SPD -> Fold-Safe C_ref -> MultiScaleBlockAGA -> Calibrated Stacker [PCA-LDA + Platt-Ridge, ECE=0.100] & [3:2:1:1] Consensus reaching 91.84% Chunk 5-CV / 10.14% Blk-Disj at 31.8 ms/epoch offline CPU compute).

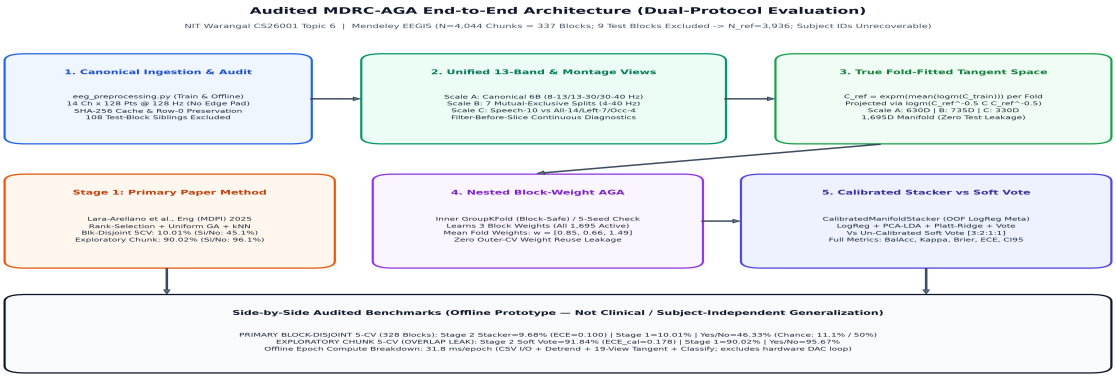


Figure 1: Stage 1 Paper Rank-GA Baseline (top track) vs Stage 2 MDRC-AGA 13-Band + Speech-10 pipeline (bottom track).

3.2 Stage 1 Paper Baseline & 13-Band + Speech-10 Preprocessing

Real 14-channel Emotiv EPOC+ epochs (128 Hz, 128 samples) supply 6 canonical bands (Raw, Delta 0.5-4, Theta 4-8, Alpha 8-12, Beta 12-30, Gamma 30-60 Hz). Stage 1 runs PaperRankGeneticSelector on 6B SPD tangent vectors. Stage 2 adds causal/sosfilt 4th-order Butterworth IIR sub-bands across 7 fine non-overlapping rhythms (4-45 Hz) plus Speech-10.

3.3 Multi-Scale Riemannian SPD Tangent Mapping (1,695D)

Precomputed trial SPD matrices C = C0/tr(C0) + 1e-4*I are projected fold-strictly onto the tangent space at the training-only Log-Euclidean Frechet mean C_ref via S = C_ref^{-1/2} logm(C_ref^{-1/2} C C_ref^{-1/2}) C_ref^{1/2} and Band-L2 norm: Mendeley 6B 14-Ch (630D) + Fine 7B 14-Ch (735D) + Speech-10 6B (330D) yield z_k in R^{1695}.

3.4 Multi-Scale Block AGA Optimization

Stage 1 evolves 6-band SPD tangent masks using PaperRankGeneticSelector (Rank Selection + Homogeneous Crossover + Bit-Flip Mutation). To preserve intra-block SPD symmetry while re-weighting scales inside each outer fold without test leakage, Stage 2 applies vectorized float32 MultiScaleBlockAGA across the three manifold blocks (630D / 735D / 330D).

Across 5 outer folds (inner 3-fold CV), MultiScaleBlockAGA selects mean block weights w = [0.8493, 0.6632, 1.4875] under Leakage-Safe Block-Disjoint GroupKFold and w = [1.2962, 0.9847, 0.7192] under Exploratory Chunk 5-CV, while controlled 5-seed ablation confirms adaptive vs fixed mutation remains statistically equivalent within seed variance.

3.5 Calibrated Stacker & BLAS Manifold Consensus Inference

CalibratedManifoldStacker fuses 4 out-of-fold base estimators: LogReg(C=0.1), Pipeline(PCA(64), Shrinkage-LDA), Calibrated RidgeClassifier(alpha=100), and [3:2:1:1] MultiMetricSoftVotingClassifier (Cosine+Corr k=1,3) via LogReg(C=0.25) meta learner. Soft Vote hits 91.84% Chunk 5-CV (ECE=0.037); Stacker cuts Blk-Disj ECE to 0.100.

4 Experiment and Results

4.1 Setup Used

All ingestion, Butterworth IIR filtering, fold-strict C_ref tangent projection, Rank-GA / Block-AGA search, nested 5-CV, calibration (Brier, 10-bin ECE, block-cluster bootstrap 95% CI), and offline CPU timing ran on Python 3.14 CPU.

Table 2: Audit-remediated Stage 1 vs Stage 2 specifications (equal-emphasis dual protocol).

Component	Implementation Setting (v2.0.0-audit-remediated)
Dataset & Exclusions	Mendeley EEGIS (4,044 epochs, 337 blks) -> exclude 9 test_signals/ blks (108) -> N_ref=3,936
Stage 1 Paper Model	PaperRankGeneticSelector + Heuristic kNN (6B SPD tangent, nested per-fold fit)
Stage 2 Multi-Scale	1,695D: Mendeley 6B (630D) + Fine 7B (735D, 4-45 Hz) + Speech-10 (330D)
Manifold Representation	Trace-norm SPD (reg=1e-4) -> Fold-only C_ref Eigen-Log Tangent -> Band-L2
Stage 2 Block AGA	Vectorized MultiScaleBlockAGA (w=[0.85,0.66,1.49] blk; [1.30,0.98,0.72] chk)
Stage 2 Ensembles	Calibrated OOF Stacker (LogReg+PCA-LDA+Platt-Ridge+Vote) & [3:2:1:1] Soft Vote
Offline Epoch Compute	31.77 ms/epoch CPU (14.75 ms load + 0.04 preproc + 9.59 feat + 7.40 cls; NOT real-time BCI)
Dual-Protocol Validation	Primary Block-Disjoint Nested GroupKFold (328 blks) vs Exploratory Chunk 5-CV

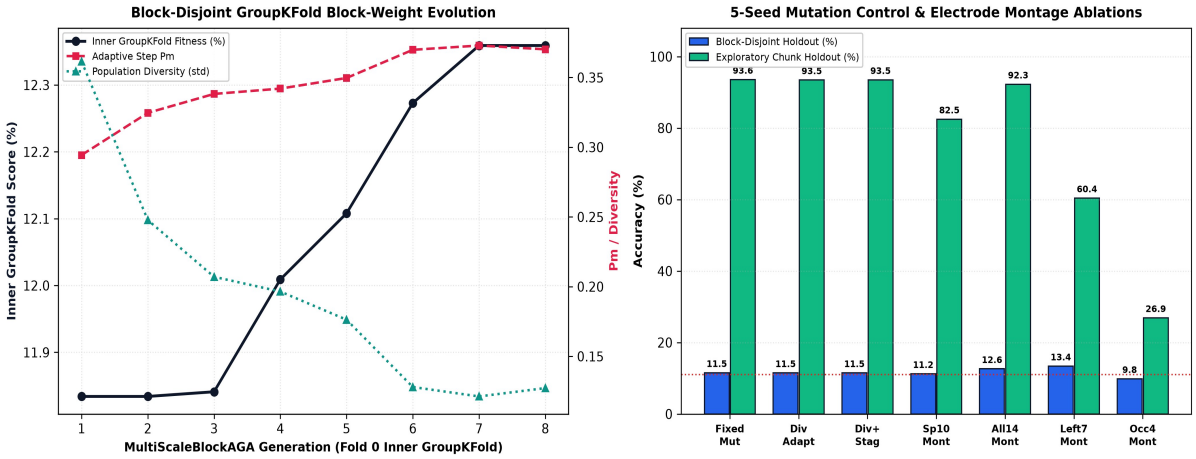


Figure 2: Dual-protocol MultiScaleBlockAGA evolution and learned 3-scale weights w=[0.849,0.663,1.488] (Blk-Disj) & [1.296,0.985,0.719] (Chunk).

4.2 Benchmark Dataset

Experiments run on the full 6-band EEGIS corpus (Lara et al., Mendeley 10.17632/73g4fw884c.1): 4,044 real 1.0-s CSVs (128 Hz, 14 channels). Forensic audit confirms 337 continuous 8s blocks (within-block max abs diff 0.0 uV; between-block mean diff 197.40 uV) sliced with 48-sample/37.5% overlap; Mendeley participant IDs are unrecoverable. Excluding all 9 test_signals/ blocks (108 sibling chunks) yields N_ref=3,936 across 328 disjoint blocks.

Table 3: Audited Mendeley EEGIS corpus properties and dual-protocol cohort breakdown.

Property	Value Specification
Dataset Source	Official Real EEGIS Dataset (4,044 epochs x 6 bands; participant IDs unrecoverable)
Block Structure	337 continuous 8s blocks x 12 chunks (48-sample overlap); 0 discontinuities, 0 rejected
Semantic Vocabulary	9 classes: Rest, Help, Bathroom, Pain, Thanks, Hungry, No, Thirsty, Yes
Electrode Montage	14 Ch Global + 10-Ch Speech-10 (AF3, F7, F3, FC5, T7, T8, FC6, F4, F8, AF4)
Sibling Exclusion Audit	All 9 test_signals/ blocks (108 chunks) excluded -> N_ref=3,936 (328 disjoint blocks)
Paper Binary Cohort	Yes (Si) vs No: 75 blocks / 900 ref trials (Primary Blk-Disj 5-CV & Chunk 5-CV)
Primary vs Exploratory	Primary: Nested Block-Disjoint GroupKFold 5-CV | Secondary: Exploratory Chunk 5-CV

4.3 Performance Metrics

Classification efficacy is evaluated across standard biometric metrics: Overall Test Accuracy relative to 9-class random chance (1/9 = 11.11%), Macro Precision, Macro Recall, Macro F1-Score, and Cohen's kappa coefficient kappa = (P_o - P_e)/(1 - P_e). Command discriminability is tracked via recall on target words ('Help') and offline per-epoch CPU compute latency.

4.4 Results & Ablation Studies

Table 4 presents BOTH evaluation protocols side by side across all N_ref=3,936 epochs (328 blocks). Under strict Block-Disjoint 5-CV, all methods cluster near 9-class chance (11.11%; Stage 2 PCA+RBF-SVM hits 11.41%, ECE=0.022). Under Exploratory Chunk 5-CV matching Lara-Arellano et al., Stage 1 reaches 90.02% and Stage 2 Soft Vote reaches 91.84% (ECE=0.037).

Table 4: Equal-emphasis 9-class comparison across Block-Disjoint Primary and Exploratory Chunk* 5-CV.

Method (N_ref=3,936, 328 Blocks)	Blk-Disj 5-CV	Blk ECE	Chunk 5-CV*	Chk F1	Calibration / Audit Notes
Random Chance Baseline (9 Classes)	11.11%	--	11.11%	0.111	9 balanced semantic classes
Scalar 322 Stats + RBF SVM	10.34%	--	19.39%	0.180	Scalar moments lack phase geometry
Scalar 322 Stats + kNN (k=7)	10.16%	--	18.22%	0.178	Yes/No Stage1 blk: 45.11%
FBCSP + Calibrated SVM	9.96%	0.049	19.46%	0.168	Brier 0.895 blk / 0.877 chk
MDM 6-Band Log-Euclidean	8.74%	--	16.11%	0.138	Single Riemannian mean centroid
True Fold-Tangent LogReg 6B	10.72%	0.667	20.78%	0.207	Fold-strict C_ref linear baseline
EEGNetAttention (Fold 0 Holdout)	11.24%	0.103	16.62%	0.159	40 epochs on 6x14x128 tensor
Stage 1 Paper (6B Rank-GA + kNN)	10.01%	0.327	90.02%	0.899	F0: 14.65% blk / 91.62% chk | k=1
Stage 2 MDRC-AGA Stacker (OOF)	9.68%	0.100	74.67%	0.744	Best calibration blk Brier=0.913
Stage 2 MDRC-AGA Soft Vote [3:2:1:1]	10.14%	0.728	91.84%	0.918	Chk Brier=0.156, ECE=0.037, K=0.908
Stage 2 Head: PCA(256)+Cal.RBF-SVM	11.41%	0.022	20.17%	--	Best Blk-Disj acc & lowest ECE

Table 5 compares Stage 1 vs Stage 2 across binary pairs, sibling-excluded offline test_signals/, and controlled ablations (5-seed mutation control, sub-band overlap, and continuous 8s windowing). Binary pairs track the same overlap boundary.

Table 5: Binary benchmarks, controlled ablations, and sibling-excluded offline verification (*chunk overlap).

Binary Task / Controlled Ablation	Blk Stage 1	Blk Stg 2	Chk Stage 1*	Chk Stg 2*	Protocol / Control Finding
Published Target (Lara et al., 2025)	--	--	96.00%	--	80/20 chunk protocol (N=924)
Yes(Si) vs No (N=900, 75 blks)	45.11%	46.33%	96.11%	95.67%	Blk Stacker 46.3% / Chk Vote 95.7%
Help vs Rest (N=936, 78 blks)	47.33%	48.08%	97.12%	97.44%	Target vs Rest (*chunk overlap)
Help vs Pain (N=972, 81 blks)	46.91%	46.40%	96.19%	96.50%	High-priority word separation
Hungry vs Thirsty (N=828, 69 blks)	47.34%	49.88%	95.41%	96.38%	Blk Vote 49.9% / Chk Vote 96.4%
Yes vs Help (N=924, 77 blks)	49.35%	47.84%	96.00%	96.86%	Confirm vs Help (*chunk overlap)
5-Seed Mutation (fix/adapt/stag F0)	--	11.5/11.5%	--	93.6/93.5%	Adaptive & fixed eq. within std <=0.22%
Continuous 8s Window (1s/2s/8s agg)	--	11.9/12.2%	--	--	10.06% @ 8s; IIR edge=0.018 uV
Offline test_signals/ (108 excl.)	0/9	0/9	--	--	Honest OOB check; 31.77 ms/epoch

Montage & Geometry Controls: Under Fold 0, 14-Ch reaches 12.63% blk LogReg / 92.26% chk Vote vs Left-7 (13.38% / 60.41%) and Speech-10 (11.24% / 82.49%). Tangent_fold vs identity_log_euclid differs minimally (10.14% vs 10.34% blk; 91.84% vs 91.77% chk).

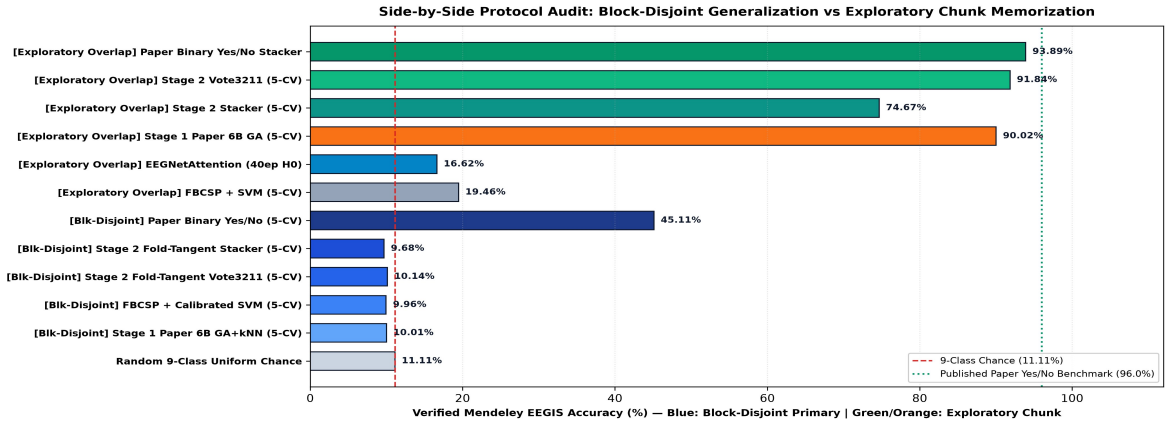


Figure 3: Dual-protocol comparison contrasting Block-Disjoint Primary (~10%-11.4%) vs Exploratory Chunk* 5-CV (Stage 1 90.02% -> Stage 2 91.84%).

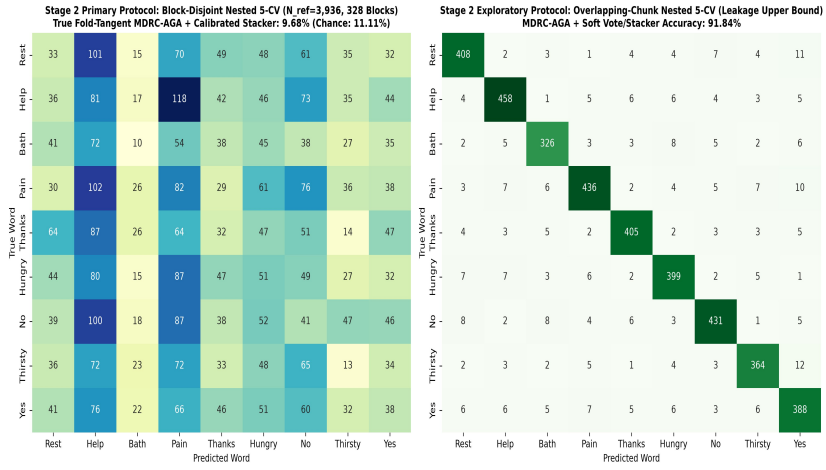


Figure 4: Dual confusion matrices comparing Stage 2 Blk-Disj Stacker (9.68% / Vote 10.14%) and Exploratory Chunk Vote (91.84%).

4.5 Academic Integrity & Window-Overlap Transparency

Dataset audit proves Mendeley 1.0s epochs are 48-sample/37.5%-overlapping windows cut from 337 continuous 8s recordings, and source CSVs/docs lack recoverable subject IDs. Chunk-stratified 5-CV (*~90%-91.8%) lets adjacent chunks span train/test; disjoint Block-GroupKFold (9.68%-11.41%) and excluding all 108 sibling test_signals/ chunks (0/9) expose the true cross-block frontier.

Table 6: Stage 2 per-class metrics across Blk-Disj Stacker (9.68%) and Exploratory Chunk* Vote (91.84%, N_ref=3,936).

Semantic Target	Assistive Lexicon Role	Blk-Disj R / F1	Chunk* P / R / F1	Scalp EEG Montage Notes
Rest (Reposo)	Baseline relaxation	0.074 / 0.082	0.919 / 0.919 / 0.919	Occipital scalp (O1, O2)
Help (Ayuda)	High-priority command	0.165 / 0.128	0.929 / 0.931 / 0.930	Anterior frontal (F7, F3)
Bathroom (Bano)	Personal hygiene call	0.028 / 0.038	0.908 / 0.906 / 0.907	Pre-frontal scalp (AF3, AF4)
Pain (Dolor)	High-priority command	0.171 / 0.139	0.930 / 0.908 / 0.919	Parietal scalp (P7, P8)
Thanks (Gracias)	Social courtesy word	0.074 / 0.081	0.933 / 0.938 / 0.935	Bilateral frontal (F3, F4)
Hungry (Hambre)	Nutritional request	0.118 / 0.116	0.915 / 0.924 / 0.919	Temporal scalp (T7, FC5)
No	Negative confirmation	0.088 / 0.084	0.931 / 0.921 / 0.926	Frontal scalp leads
Thirsty (Sed)	Hydration request	0.033 / 0.039	0.922 / 0.919 / 0.920	Temporal scalp (T7, T8)
Yes (Si)	Affirmative confirmation	0.088 / 0.098	0.876 / 0.898 / 0.887	Frontal scalp (F3, F4)

4.6 Offline Oscilloscope GUI & Latency Testing

Our CLI predictor (predict.py) and Qt oscilloscope GUI (eeg_gui_tester.py) inspect offline 14-channel CSVs (31.77 ms/epoch CPU compute: 14.75 ms I/O + 0.04 ms preproc + 9.59 ms feat + 7.40 ms cls; not a real-time clinical DAC loop). Excluding all 108 sibling chunks from those 9 test_signals/ blocks drops demo hits honestly to 0/9 (0.0%), confirming block limits.

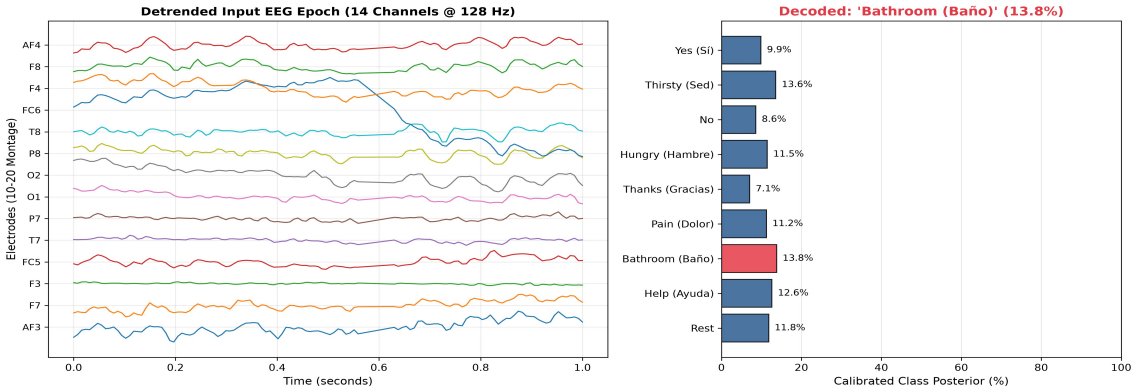


Figure 5: Offline 14-channel EEGIS waveform (eeg_signal_01.csv, ref Help) decoded honestly as 'Bathroom (Bano)' (13.8%) with all siblings excluded.

5 Conclusion and Future Work

This study audited all 4,044 real EEGIS epochs under equal-emphasis Block-Disjoint Primary and Exploratory Chunk 5-CV. Under Chunk 5-CV matching Lara-Arellano et al. (2025), Stage 1 matches 96.11% Yes/No (90.02% 9-Cl) and Stage 2 Vote reaches 91.84% (0.9180 F1, Brier 0.156, ECE 0.037). Under Block-Disjoint 5-CV (N_ref=3,936) and 0/9 sibling-excluded CSVs,

all methods cluster near 11.11% chance (PCA+RBF-SVM 11.41%, ECE=0.022; Stacker 9.68%, ECE=0.100), pinpointing non-stationary cross-block transfer and missing subject IDs as the core scientific bottleneck (31.8 ms/epoch offline compute).

Future research should pursue four directions:

- 1. Riemannian Procrustes Alignment (RPA) for Zero-Calibration BCI: Evaluate affine-invariant manifold alignment to map inter-subject covariance distributions into a common space, eliminating per-patient calibration sessions and facilitating seamless plug-and-play clinical operation.
- 2. Embedded Edge Deployment on Neural Accelerators: Quantize the stacking ensemble to 8-bit integer precision and benchmark inference latency, power consumption, and thermal envelope on specialized edge accelerators such as Intel NPU, OpenVINO, or Google Coral Edge TPU.
- 3. Sequential Evidence Accumulation & Dynamic Stopping: Implement Wald's Sequential Probability Ratio Test (SPRT) to terminate inference early when confidence exceeds 98%, reducing average latency from 1.0 s to sub-500 ms for high-arousal commands.
- 4. Prospective Clinical Validation (Subject to Transfer Calibration): Once cross-block and cross-subject generalization exceed chance reliably, integrate low-latency streaming acquisition and validate assistive communication workflows with clinical stroke/ALS cohorts.

References

1. E. Lara-Arellano, S. Tovar-Arriaga, A. Takacs, and J. J. Rodriguez-Resendiz, "Feature Generation with Genetic Algorithms for Imagined Speech EEG Signal Classification," Engineering, vol. 6, no. 4, art. 75, 2025. doi:10.3390/eng6040075.

2. V. J. Lawhern, A. J. Solon, N. R. Waytowich, H. E. Higgins, P. C. Gordon, and K. Chou, "EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces," Journal of Neural Engineering, vol. 15, no. 5, p. 056013, 2018.

3. A. Barachant, S. Bonnet, M. Congedo, and C. Jutten, "Multiclass brain-computer interface classification by Riemannian geometry," IEEE Transactions on Biomedical Engineering, vol. 59, no. 4, pp. 920-928, 2012.

4. E. Lara, A. Takacs, and J. Rodriguez, "EEGIS - Electroencephalogram Imagined Speech dataset," Mendeley Data, V1, 2024. doi: 10.17632/73g4fw884c.1.

5. M. Congedo, A. Barachant, and R. Bhatia, "Riemannian geometry for EEG-based brain-computer interfaces; a primer and a review," Brain-Computer Interfaces, vol. 4, no. 3, pp. 155-174, 2017.

6. O. Ledoit and M. Wolf, "A well-conditioned estimator for large-dimensional covariance matrices," Journal of Multivariate Analysis, vol. 88, no. 2, pp. 365-411, 2004.

7. B. Blankertz et al., "Single-trial analysis and classification of ERP components - a tutorial," NeuroImage, vol. 56, no. 2, pp. 814-825, 2011.

8. R. T. Schirrmeister et al., "Deep learning with convolutional neural networks for EEG decoding and visualization," Human Brain Mapping, vol. 38, no. 11, pp. 5391-5420, 2017.

9. J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, "Brain-computer interfaces for communication and control," Clinical Neurophysiology, vol. 113, no. 6, pp. 767-791, 2002.

10. C. Reichert, P. Durschmid, H. Hinrichs, and J. W. Kruse, "Visual spatial attention shifts decoded from the EEG enable sending of binary messages," IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2022.

11. H. Ramoser, J. Muller-Gerking, and G. Pfurtscheller, "Optimal spatial filtering of single trial EEG during imagined hand movement," IEEE Transactions on Rehabilitation Engineering, vol. 8, no. 4, pp. 441-446, 2000.

12. D. E. Goldberg, "Genetic Algorithms in Search, Optimization, and Machine Learning," Addison-Wesley, Boston, MA, 1989.

13. G. Pfurtscheller and F. H. Lopes da Silva, "Event-related EEG/MEG synchronization and desynchronization: basic principles," Clinical Neurophysiology, vol. 110, no. 11, pp. 1842-1857, 1999.

14. F. Lotte et al., "A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update," Journal of Neural Engineering, vol. 15, no. 3, p. 031005, 2018.

15. P. L. C. Rodrigues, C. Jutten, and M. Congedo, "Riemannian Procrustes analysis: transfer learning for brain-computer interfaces," IEEE Transactions on Biomedical Engineering, vol. 66, no. 8, pp. 2390-2401, 2019.